USN | | | | | | | | | | | [22ESCS11]

FIRST SEMESTER B.Tech EXAMINATION, DECEMBER 2022

PROBLEM SOLVING THROUGH PROGRAMMING IN C

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

PART-A (5X2=10)

ANSWER ALL QUESTIONS

1.	**a.**	Distinguish recursion and iteration.	L- 2	CO1	**02**
	b.	Correlate else if ladder and switch case statement	L- 2	CO2	**02**
	c.	Explain the significance of Null character in string.	L- 2	CO3	**02**
	d.	Differentiate **while** and **do-while** loops.	L- 2	CO4	**02**
	e.	Difference between Static Memory Allocation and Dynamic Memory Allocation.	L- 2	CO5	**02**

PART-B (6X5=30)

2.	**a.**	Discuss the major components of a digital computer with suitable block diagram. Also discuss the function of each component.	L- 2	CO1	**05**
		OR			
	b.	What are operators? Mention different types of operators in C. Explain the difference between operator precedence and associativity with suitable example.	L- 2	CO1	**05**
3.	**a.**	Write a program to print the pattern 1 12 123 1234	L- 2	CO2	**05**
		OR			
	b.	Write a program to print the pattern 1 2 3 4 5 6 7 8 9 10	L- 2	CO2	**05**
4.	**a.**	Write a program to find out the greatest number out of three numbers.	L- 3	CO3	**05**
		OR			
	b.	Write a program to find the sum of first 10 elements of Fibonacci series using function. [Fibonacci Series are the numbers in the following integer sequence:- 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...].	L- 3	CO3	**05**
5.	**a.**	Explain the use of **break** and **continue** statements with a suitable example.	L- 2	CO4	**05**
		OR			
	b.	Explain **switch** Case statement with an example.	L- 2	CO4	**05**

6.	**a.**	#include<stdio.h>
		void main()
		{
		int a[5] = {5, 1, 15, 20, 25};
		int i, j, m;
		i = ++a[1];
		j = a[1]++;
		m = a[i++];
		printf("%d, %d, %d", i, j, m);
		}
		Interpret the output.
			L- 3	CO5	**05**

		OR			

		#include <stdio.h>
		int main(void)
		{
		char p;
		char buf[10] = {1, 2, 3, 4, 5, 6, 9, 8};
		p = (buf + 1)[5];
		printf("%d", p);
		return 0;
		}
		Interpret the output.
			L- 3	CO5	**05**

7.	**a.**	void main() {
		int a[] = {1,2,3,4,5}, *p;
		p = a;
		++*p;
		printf("%d ", *p);
		p += 2;
		printf("%d ", *p);
		}
		Interpret the output.
			L- 3	CO6	**05**

		OR			

	b.	#include <stdio.h>
		void foo(int*);
		int main()
		{
		int i = 10;
		foo((&i)++);
		}
		void foo(int *p)
			L- 3	CO6	**05**

		{
		printf("%d\n", *p);
		}
		Interpret the output.			

PART-C (6X10=60)

8.	**a.**	Write a program to swap the two numbers through Call by Reference.	L- 3	CO1	**10**
		OR			
	b.	Write a program to find the factorial of a number using recursive function.	L- 2	CO1	**10**
9.	**a.**	Take the three-digit number from the user then write a program to check whether the entered number is a palindrome or not.	L- 3	CO2	**10**
		OR			
	b.	Write a program to search for an element using Linear search and explain with an example.	L- 3	CO2	**10**
10.	**a.**	Explain Bubble sort with example.	L- 3	CO3	**10**
		OR			
	b.	Write the advantages of using array. Write a program for Matrix Addition.	L- 3	CO3	**10**
11.	**a.**	Brief the storage classes in C with proper example.	L- 2	CO4	**10**
		OR			

	b.	#include<stdio.h>
		int main()
		{
		int a;
		char *x;
		x = (char *) &a;
		a = 512;
		x[0] = 1;
		x[1] = 2;
		printf("%d\n",a);
		return 0;
		}
		Interpret the output of above program
		i. Machine dependent
		ii. 513
		iii. 258
		iv. Compiler Error
			L- 3	CO4	**10**

12.	**a.**	Explain the difference between Call by Value and Call by Reference with proper examples. Also,define Formal Parameters and Actual Parameters.	L- 2	CO5	**10**

		OR			
	b.	Explain Dynamic memory Allocation concept with proper example.	L- 3	CO5	**10**
13.	**a.**	Explain the different type of modes and I/O functions in file handling.	L- 3	CO6	**10**
		OR			
	b.	Explain File Handling and write a program for copying the content of the one file to another file.	L- 3	CO6	**10**